Woojin Cho

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WoojinCho-Ryan

Seoul, South Korea

Publications

- **K-WebArena: Evaluating Web Navigation Agents on Korean Websites** *Preprint*
  - **Woojin Cho**, Dongkeun Yoon, Nahyun Lee, Guijin Son, Eunsu Kim, Jaewon Cho, Haneul Yoo, Jaeyeon Kim, Seungone Kim
- **How Do LLMs Change Predictions Under Negation?** *Preprint*
  - Jongwook Yoon, Jongwon Lim, Sungjib Lim, **Woojin Cho**, Yohan Jo

Education

- **Seoul National University** *Mar 2024 – Present*
  B.S. in Computer Science and Engineering (Minor in Mathematical Sciences) Seoul, South Korea
  - **GPA: 4.06 / 4.30**, **Major GPA: 4.20 / 4.30**
  - Expected Graduation: Aug 2029 (Mandatory Military Service: Jan 2027 – Jun 2028)
- **Hansung Science High School** *Mar 2021 – Feb 2024*
  Specialized in Mathematics and Science Seoul, South Korea

Relevant Coursework

- **Computer Science:** Advanced Algorithms, Data Structures, Discrete Mathematics, Logic Design
- **Mathematics:** Linear Algebra I, Analysis I, Differential Equations, Abstract Algebra I

Research Interests

- **Primary:** Mechanistic Interpretability, Autoformalization, Agentic Benchmarks

Research Experience

- **Research Intern Mentorship** *Feb 2026 – Oct 2026*
  Mentor: Seungone Kim
  - **Project: Korean Web Navigation Benchmark**
  - **Korean Web Environment Simulation:** Constructed reproducible Docker-based web environments for major Korean services (e.g., Naver, KTX) to enable realistic evaluation of web navigation agents.
  - **Benchmark Localization:** Curated a diverse set of Korea-specific navigation tasks and queries, addressing the cultural and linguistic gaps in existing English-centric benchmarks like WebArena.
  - **Framework Adaptation & Scalability:** Adapted the WebArena framework to handle Korean web interfaces and designed the architecture to support future extensions into Computer Use Agents (CUA).
- **Language & Data Intelligence Lab, Seoul National University** *Sep 2025 – Jan 2026*
  Undergraduate Research Intern (Advisor: Prof. Seung-won Hwang) Seoul, South Korea
  - **Project: Autoformalization & Structured Reasoning**
  - **Designing Intermediate Representations:** Designed a structured JSON schema to bridge the semantic gap between ambiguous natural language math problems and formal logic (Lean 4), enabling rigorous verification.
  - **Data Pipeline Engineering:** Built a robust preprocessing pipeline to convert unstructured mathematical texts into structured evaluation datasets, handling diverse edge cases in logical statements.
  - **Hallucination Analysis:** Analyzed failure modes of LLMs in reasoning tasks, identifying that enforcing output structure significantly mitigates logical hallucinations.

PROJECTS

- **Adaptive Robustness of CountSketch: Collision Noise & Superposition** *Spring 2026*
  Theory, Randomized Algorithms, Mechanistic Interpretability
  - Proved a complete non-adaptive analysis of CountSketch — unbiasedness, variance bound, and median amplification — isolating the single independence hypothesis each proof relies on.
  - Derived a **conditional-expectation lemma** showing that adaptive query selection converts mean-zero collision noise into biased noise, connecting the mechanism to the Hardt–Woodruff impossibility theorem.
  - Proposed a **formal dictionary** between CountSketch and toy models of superposition, framing adaptive fragility as a precise toy model of an interpretability hazard in compressed neural representations.
- **Real-time Visual Perception & Perspective Correction** *Spring 2024*
  Python, NumPy, Linear Algebra
  - Implemented a **real-time vision algorithm** to rectify distorted monitor screens (e.g., trapezoids) into canonical rectangular views using **Homography** matrix operations.
  - Applied **Linear Algebra** principles to map pixel coordinates from a tilted camera feed to a target 2D plane, correcting geometric distortions.
  - Optimized the pipeline for video streams, developing robustness against camera jitter and motion blur to ensure stable detection.

SKILLS

- **Languages:** Python (Advanced), C++, Java, Lean 4, Coq
- **Frameworks & Libraries:** PyTorch, Hugging Face Transformers, NumPy, Pandas
- **Tools & DevOps:** GitHub, Linux (Ubuntu), LaTeX